

Managed ArgoCD Scalability

Slide Content Document

Slide 1: Managed ArgoCD Scalability

Expanding GitOps as a Platform Capability

Objective

Evaluate how Managed ArgoCD can evolve to support broader platform adoption while maintaining a scalable and reliable operating model.

Agenda

1. ArgoCD Overview and Key Capabilities
 2. Current Managed ArgoCD Adoption
 3. Growth Opportunities
 4. Problem Statement and Impact Analysis
 5. Proposed Architecture Direction
 6. Feedback and Next Steps
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Slide 2: What is ArgoCD?

GitOps Control Plane for Kubernetes

Key Capabilities

- Git as the Source of Truth
- Continuous Reconciliation
- Multi-Cluster Management
- Application Health & Deployment Visibility
- ApplicationSets for Fleet-Wide Deployments
- Fine-Grained RBAC & Multi-Tenancy

Why It Matters

Provides a standardized way to deploy, manage, and monitor Kubernetes workloads at scale.

Slide 3: Managed ArgoCD Adoption Today

Proven in Production

Platform Services

Managed Spark Streaming

- Namespace-based platform offering
- Platform-owned controllers and infrastructure managed through ArgoCD
- GitOps-driven operational model

Managed PostgreSQL

- Database-as-a-Service platform
- Operators, controllers, and supporting infrastructure managed through ArgoCD
- Consistent deployment and lifecycle management

Application Teams

Examples:

- API Proxy
- P13
- Additional single-tenant platform consumers

Value Delivered

- Standardized deployment workflows
 - Operational visibility
 - Automated reconciliation
 - Fine-grained RBAC and access control
 - Consistent GitOps operating model
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Slide 4: Why Expand Managed ArgoCD?

Opportunity for Broader Adoption

Many platform offerings share similar characteristics:

- Namespace-based service offerings
- Large cluster fleets
- Shared infrastructure components
- Platform-managed controllers and operators

How ArgoCD Can Help

- GitOps deployment workflows

- Fleet-wide infrastructure management
- Standardized operational patterns
- Application health and deployment visibility
- Fine-grained access control and self-service capabilities

Opportunity

Expand Managed ArgoCD into a reusable platform capability that can be leveraged across multiple platform offerings.

Slide 5: Problem Statement & Quantification

Current Architecture Limits Future Adoption

Current Architecture

All Managed ArgoCD consumers depend on a single ArgoCD control plane.

Current Scale

Environment	Clusters	Applications
Production	~100	~1,100
Non-Production	~150	~2,000
Total	~250	~3,100

Current Scaling Status

Managed ArgoCD has already been scaled to a 14-node cluster.

Broader Adoption Opportunity

Metric	Estimated Scale
Clusters	2,000+
Applications	200,000+

Problem Statement

- Shared operational dependency
- Increased blast radius
- Centralized scaling concerns
- Increased operational risk

Key Observation

The challenge is enabling Managed ArgoCD to evolve into a broader platform capability without being constrained by a single control plane architecture.

Slide 6: Solution Requirements & Value Proposition

What Must the Next Architecture Achieve?

Requirements

- Scalability
- Reliability
- Operational Simplicity
- Unified Experience

Value Proposition

Enable Managed ArgoCD to evolve into a reusable GitOps platform capability that can be adopted across multiple platform offerings.

Success Criteria

- Enable broader platform adoption
 - Maintain reliability
 - Preserve user experience
 - Support long-term growth
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Slide 7: Proposed Architecture Direction

Distributed Control Plane Architecture

High-Level Approach

Introduce multiple ArgoCD control planes behind a unified Managed ArgoCD experience.

Architectural Principles

- Horizontal Scalability
- Reduced Blast Radius
- Operational Independence
- Unified Consumer Experience

Expected Outcome

A scalable foundation that enables broader adoption while preserving native ArgoCD capabilities.

Slide 8: Next Steps & Feedback Requested

Checkpoint 2

- Placement Strategy
- Routing Architecture
- Capacity Planning
- Migration Strategy
- Operational Model
- Reliability & DR

Feedback Requested

- Problem Statement Validation
- Architecture Direction Validation
- Additional Risks and Constraints

Decision Requested

Alignment to proceed with detailed design exploration.